

Certificate of completion

Core Training - Simulator-based

Certificate number: E202309141970810

Robot type: e-Series

Julio Antonio Caballero Mora

14. September 2023



Hector Salazar Rojas
Universal Robots Academy Mexico



Curriculum

Julio Antonio Caballero Mora 14. September 2023

Module 1: Pick-and-place application

- Moving the robot by using the "Move" tab
- Knowledge about the I/O Connections and how-to setup
- How to configure the TCP
- How to setup a Home Position
- Basic knowledge about URCaps
- Apply skills acquired in the online training to an actual robot
 - Programming a Pick-and-place application

Module 2: Safety settings

- Configuration and use the available safety settings
 - Robot limits
 - Joint limits
 - Safety planes
 - Tool position
 - Direction
 - Configurable Inputs/Outputs
 - Safe Home

Module 3: Optimize a pick-and-place application

- Basic knowledge about singularities
- The different movement types and how to use them
 - MoveJ
 - MoveL
 - MoveP
 - MoveC
- How to configure a blend radius
- Configuration of speed and acceleration of movements and individual waypoints
- How to use the Time to investigate the cycle time
- Creation of a clear program structure

Module 4: Easy startup

- Configuration of a default program that starts automatically when the robot powers up
 - Programming a BeforeStart-Sequence
 - How to program relative Waypoints
 - How to setup the remote control
 - Setup a default program

Module 5: Program flow

- Configuration and use if-else commands
- Basic knowledge about variables
- How to program a subprogram

Module 6: Palletizing

- Configuration and usage of the palletizing template

Module 7: Force control (simple)

- Knowledge about the Force control (simple)
- Knowledge about and how to program Threads
- Configuration of the Force control (simple)
- Using the data of the integrated force and torque sensor in a program

Module 8: Process application with operator selection

- Configuration of the TCP, orientation and payload using the available wizards
- Programming loop statements
- Prompt operator for variable input
- Configuration and usage of switch-case commands
- Catching and handling an invalid entry

Module 9: Flexible redeployment

- How to create a coordinate system (Plane)
- How to program relative to a coordinate system

Module 10: Implementation plan

- A structured method (ten steps) to identify and evaluate the complexity of the automation capabilities in your cobot

Module 11: Online resources

- Free software updates
- User, service and script manuals
- CAD data of the robot, Teach Pendant and controller
- Offline simulator

Quiz

- Completion of the UR Academy Quiz Passing score of 95.05



UNIVERSAL ROBOTS

A Teradyne Robotics Company